



A multimodal framework for the evaluation of patients' weaknesses, supporting the design of customised AAL solutions

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ARTICLE INFO

Keywords:

Expert systems
Computer vision
Real-time
Ambient-assisted living

ABSTRACT

The recovery of motion abilities after a physical or neurological trauma is a long and winding road that is usually supported by medical staff. In particular, occupational therapists (OTs) play a fundamental role in assessing the performances of patients in daily life tasks and suggesting better practices and aids. The goal of OTs is to promote the patients' abilities, fostering their independence with the minimum needed supervision. To this purpose, the possibility of remotely operating while being able to monitor the subject with a sufficient level of detail is highly desirable. The development of advanced tools for patient monitoring and supervision with limited intrusiveness is a scientific challenge that spans different technological areas: sensing, localisation, tracking, measurement, business intelligence. The final goal is to provide medical doctors with a set of meaningful metrics related to the patient's activity. Among them, occupation of living spaces, motion patterns, posture, as well as fine-grained motion-related parameters (e.g., grasping objects, performing elementary tasks of variable difficulty), can be of great help for the medical staff to assess the degree of independence of the user and to recommend suitable assisting devices, more appropriate organisation of living spaces, effective rehabilitation procedures. In this work, we describe a sophisticated setup that has been developed in strict cooperation with medical experts and researchers in both engineering and medicine, to create an augmented reality physical environment supporting occupational therapists and rehabilitation staff in evaluating their patients' performance and therapy activities. The proposed system is fully automatic and the users detection and tracking is completely markerless and vision-based. It comprises a set of realistic highly-structured living environments, including sensing and actuation devices of different types, which are thoroughly described in Tables 1 and 2. The collected data are analysed by an automated expert system to extract the requested information, which is then presented inside the user interface. The underlying technological complexity is hidden to both patients and the medical staff, to guarantee minimum intrusiveness for the former and maximum operational easiness for the latter. During and after the patients staying inside the living spaces, the medical personnel is enabled to access a large variety of data through a unified interface and is provided with a pre-filled version of the standard patient card, a document that is otherwise filled in by hand. This allows both reducing the time requested for a medical diagnosis and producing a more objective and measurable assessment. A fully operating setup of our system is currently deployed within the living lab AUSILIA (Pisoni et al. 2016).

1. Introduction

Occupational therapy aims at assessing and evaluating the activities of various types of patients, in order to guide them in regaining or improving their motor or mental skills through rehabilitation. However, fostering independence during and after rehabilitation is a fundamental yet difficult process. In fact, the interaction with fragile subjects requires a special mix of technical competence and sensitivity to personal

elements, in a patient work of trust enforcement. From OTs' perspective, the possibility of observing patients performing daily activities allows understanding their weaknesses, tracing down their improvements in the rehabilitation process, and making it possible to identify the most suitable assistive technologies, tools, or clinical paths that can contribute to provide them with a *new normality*. While direct observation is currently the most adopted practice, well-established in the literature with plenty of examples (Anastasopoulos, Niebuhr, Bartelt,

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<https://doi.org/10.1016/j.eswa.2022.117172>

Received 12 May 2021; Received in revised form 21 December 2021; Accepted 31 March 2022

Available online 12 April 2022

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Koch, & Rausch, 2005; Clarke, Marshall, Black, & Colantonio, 2002; Comas-Herrera et al., 2006; Fuglsang, Hansen, Mergel, & Röhnebak, 2021; Koch, Wottrich, & Holmqvist, 1998), it typically introduce a bias in the analysis, because of the physical presence of the therapist together with the patient. With this work we aim at mitigating this bias as much as possible, by providing the medical staff with a sophisticated environment that allows monitoring the patients in completely realistic situations and in their typical living conditions (alone or supported by a caregiver). To this purpose, the proposed system allows monitoring the person both in real-time and offline, and generates an augmented presentation of the captured information by automatically processing the sensor data, analysing them, generating enhanced views, annotating, overlaying measures and data, and so on. The same data are also structured in reports for better accessibility.

The main goals of the proposed system can be summarised as follows:

- Enriched data and metrics: the system provides a number of information on the observed environment, coming from both the direct sensor observation (cameras, 3D sensors, environment/object sensors, wearable, etc.) and the integrated processing of such data with a set of tools (motion analysis, tracking, localisation, action patterns, space occupation, etc.)
- Fully automatic analysis: the implemented architecture is designed to meet the requirements of non-expert users (OTs' and medical staff's). The user has just to turn-on the system and start collecting trials and analytics. Afterwards, he can access the data through a user-friendly multi-modal interface, to infer the clinical conditions of the patients and assess their progresses.
- Minimum invasiveness: the configuration of the environment is organised in order to hide as much as possible the complexity of the infrastructure. Sensors are embedded into the environment and in objects and equipment placed in remote, so that the user can live a 'feel-at-home' experience.
- Portability and real-time compliance: the system is organised according to a distributed and modular structure, which allows to dynamically configure new environments and port it to other scenarios, upon need. Furthermore, it allows the involved staff to access and visualise the metrics and results in real-time from mobile devices such as tablets and laptops, without waiting for additional post-processing steps.

2. Related work

Activity and behaviour monitoring of patients affected by cognitive and/or physical diseases is reported in several studies, and could be performed in both clinical settings and daily living environments for different purposes. As far as clinical monitoring is concerned, the literature shows that Intensive Care Units (ICU) are among the most critical hospital wards where an accurate control of activities is of paramount importance. In this case, monitoring is performed mainly using depth cameras and other non-invasive devices (Davoudi et al., 2019; Ma et al., 2017). Other studies proposed ambient monitoring systems to assess the fulfilment of safety procedures and to supervise complex equipment (Chen, Cremer, Zarei, Segre, & Polgreen, 2015).

An interesting application of in-hospital monitoring concerns patients' evaluation for the elderly or subjects affected by chronic diseases. In this case, technologies support the clinical staff in both assisting the patients and assessing their needs, with the aim of determining whether they can return to normal life and the possible assignment of human support or aids (Butaslac et al., 2020; Stocco et al., 2019). The monitoring system is requested to acquire manifold information, including physiological parameters, level of stress, behaviours, but also the ability to perform daily activities. Given the different nature of each disability and their diverse impact on the capability in performing various activities, customisation is a key issue. Various solutions have been

reported in literature, mostly based on the adoption of sensor networks. On one side, wearable sensors are effective in measuring physiological parameters (ECG, body temperature, blood pressure Chen, Lee, Chen, Huang, & Luo, 2010), collected while the subject performs daily activities. For instance, MyHeart (Pacelli, Loriga, Taccini, & Paradiso, 2006) and MagIC (Di Rienzo et al., 2005) proposed wearable monitoring systems using small sensors attached to everyday clothes, to ensure unobtrusive recording of cardio-respiratory signals or user's movements. On the other side, environmental sensors can collect useful information with lower obtrusiveness. For instance, the use of radar or acoustic sensing devices has been proposed to detect user's activities (Chen, Zhang, Kam, & Shue, 2005). A more detailed overview of the solutions and platforms so far proposed to tackle physiological and environmental monitoring can be found in the works by Spasova et al. focusing on fall detection (Spasova & Iliev, 2014), Bonato et al. reviewing wearable sensors based solutions (Bonato, 2010) and Acampora et al. focusing on environmental sensor architecture approaches (Acampora, Cook, Rashidi, & Vasilakos, 2013).

A major limitation of monitoring systems developed so far lies in the fact that they are usually tailored to the specific needs which they are designed for. For instance, objective assessment of Parkinson disease is achieved through various specific solutions reviewed in Maetzler, Domingos, Srulijes, Ferreira, and Bloem (2013), rehabilitation accounts a wide and fast growing spectrum of customised systems, such as the recently published (Faber, Kingma, Chang, Dennerlein, & Van Dieen, 2020). On the contrary, the typical situation when dealing with elderly or disabled people is a unique combination of physical and cognitive limitations and co-morbidities that requires a more holistic approach to be fully assessed. Previous attempts to tackle this problem are reported in López, Ponce, Piccinini, Berenguer, and Roberti (2016), addressing severely injured people, and in Harmo, Taipalus, Knuuttila, Vallet, and Halme (2005) and Ortenzi et al. (2015), where different ambient-assisted living (AAL) solutions are reviewed and organised.

In this work we describe a sophisticated in-hospital smart environment designed to help medical doctors and therapists in assessing the needs of elderly and disabled people and to customise appropriate AAL solutions able to guarantee a safe and functional living environment for the patient. To this aim, we did not focus on a specific technology, but we integrated into a dedicated smart space a large network of sensors and actuators able to capture diverse and complementary information on every activity performed by the user, as well as his status and physiological parameters. The developed environment makes it possible to monitor a wide variety of patients, with different combinations of disabilities and weaknesses, in a controlled and adaptive environment. This approach allows to easily identify the needs of the patient and later to translate them into a personalised AAL solution.

3. Context: the AUSILIA facilities

The presented expert system has been developed to work inside a domotic apartment and a gym with different reconfigurable environments, which are hosted inside the Villa Rosa hospital in Pergine, Trento (Italy). The units of the AUSILIA project represent the spaces where users spend their time carrying out their daily activities. The apartment was designed to accommodate people with different cognitive or physical impairments and therefore with different needs. For such a reason, therapists ask to evaluate their patients in an apartment that allows to cover different type of situations. Given the size of the domotic apartment (~70 m²) that did not allow to meet all the therapists' requests and the patients' needs, a gym was realised as a parallel environment of ~400 m² where even the most specific solution could be tested. In the gym, environments can be highly reconfigurable such as the door clearances or corridor widths. With these facilities, therapists have the opportunity to have a real test-bench, allowing an effective understanding of their patients' problems and needs. Moreover, both the domotic apartment and the gym have been a perfect test bench

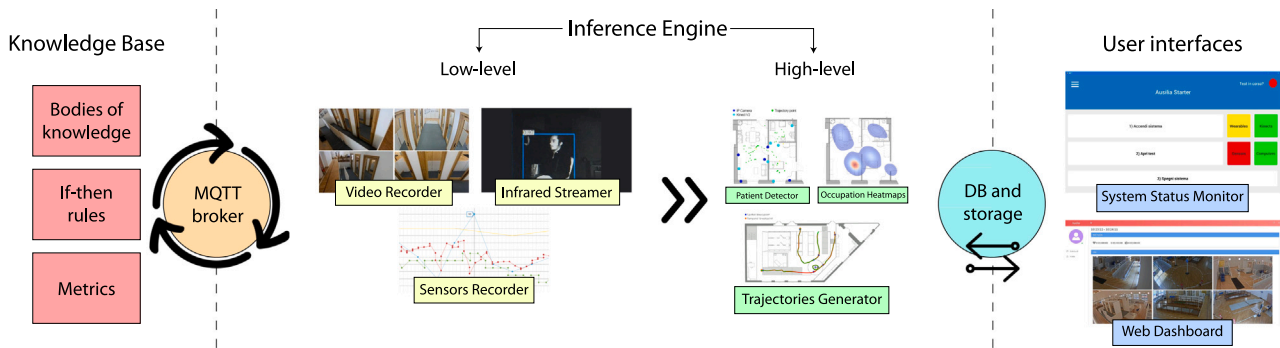


Fig. 1. An overview of the proposed expert system. The system is enabled to work with different kinds of low-level data such as video streams, environmental and wearable sensors data. The inference engine deals with transforming all the low-level information into high-level features, according to the knowledge base. An MQTT broker serves as orchestrator for all the parts of the system, while all the required features are stored into a structured database linked to a network-attached store. Finally, the end user can access and edit all the required parameters of the system through a system status monitoring console and visualise the system metrics and gathered data from an interactive web dashboard.

for two research laboratories¹² of the University of Trento in order to design, develop and test the proposed expert system in a realistic and fully-fledged AAL environment.

4. System overview

In this section we review the main building blocks of the system and the way they are integrated and orchestrated to collect and process the data. Our proposed system is depicted in Fig. 1. On the sensing side, a set of camera (IP, infrared and stereo), wearable, and environmental sensors are employed. The raw data extracted in real-time by these sensors are processed by the inference engine to obtain high-level features, according to the knowledge base, which is jointly determined by the medical and the technical staff. All the structured information, together with the system status and parameters, is stored and can be accessed by the final user through a set of human-friendly user interfaces. The means of communication between the three main layers is a Message Queue Telemetry Transport (MQTT) orchestrator and a network-attached storage for data collection.

In the following subsections we will detail the role of each communication, inference and presentation module.

4.1. Communication modules

The proposed expert system is configured as an ensemble of different inference modules that tackle different automation tasks inside the AAL environment, based on bodies of knowledge. Each module is configured as an always-on service with on-demand processing triggered by specific MQTT messages. The MQTT broker and the network-attached database and storage are the two main communication modules of the proposed system.

- MQTT broker: it manages topics for different rooms and activities and forwards messages to the proper services. The usage of MQTT as the main orchestrator of the expert system allows for a fine-grained configuration of environments and rooms through MQTT topics, as well as allowing to save computing resources by hibernating services when not needed.
- Database and storage: the proposed expert system interfaces with a MongoDB database to store all the information used by the presentation and metrics evaluation modules, such as timestamps, events and positions inside the environment. Although the system is intrinsically distributed, all the big data such as videos and sensors outputs must reside in a unique Network-Attached Storage

(NAS), which is a necessary component, especially when multiple clients are allowed to obtain access to specific data, even simultaneously. More information on the role of the database and storage for data processing and presentation can be found in Sections 6, 7.

4.2. Inference modules

The inference engine in the proposed system transforms the knowledge base into concrete instructions, both low-level and high level. The inference services of the proposed AAL expert system are configured as follows:

Low-level:

- Video Recorder: it synchronises and records IP camera streams through the RTSP protocol in real-time, aided by GPU acceleration.
- Infrared Streamer: it synchronises and locally streams infrared streams via the UDP protocol.
- Sensors Recorder: it synchronises, filters and records both wearable and environmental sensors data in real-time.

High-level:

- Patient Detector: it processes and assembles infrared streams to detect the patient and caregivers position and rooms inside the AAL environment
- Trajectories Generator: it generates and filters human poses (Cao, Simon, Wei, & Sheikh, 2017) from both infrared and RGB streams to infer and save both the patient and the caregivers trajectories
- Occupancy Heatmaps Generator: it processes the trajectories to infer a 2D occupancy map for the different areas (Fig. 9)
- Automatic Activity Annotator: it collects additional data about the activities (e.g., force sensors, pressure sensors, etc.) and links and integrates this data with the activities themselves for future presentation.

4.3. Presentation modules

The user interface modules communicate with the network-attached storage and database to obtain structured information about the system status and the past and ongoing tests in the environment. Through these interfaces, a registered user is also allowed to edit some of the parameters of the system's configuration. The presentation modules are the following:

- System Status Monitor: it allows OTs and the medical staff to easily check for the status of the ensemble of sensors and environments, enable or disable modules, as well as allowing them to start and stop new tests and automatically record the needed data

¹ DISI UniTN <https://www.disi.unitn.it>.

² DII UniTN <https://www.dii.unitn.it>.

- **Web Dashboard:** it collects all the data from the OTs tests and presents it in an organised, everywhere accessible web interface
- **Test viewer:** it allows medical staff to review tests performed and saved in the database. This functionality shows the video of the IP cameras and the audio recordings of the tests. It also allows to reproduce the registrations starting from a specific time point.
- **3D Visualiser:** this tool is dedicated to visualise the tests in a 3D immersive environment. OTs must wear augmented reality glasses and headphones to watch the recorded tests. In this context, user can interact with the environment by virtually moving inside different views around the virtual space or by reproducing a certain dataset in slow motion, fast forward or by freezing the reproduction at a specific frame. Clinicians can also interact with the output features data on a graph in order to clearly focus on a specific frame and analyse the desired features (Fig. 6).

5. Sensing subsystem

The system exploits a large sensing infrastructure to achieve evidence of all the necessary parameters related to the activities performed by the user. This includes the recording of the action itself, in 2D and 3D, the acquisition of additional information coming from the environment and the manipulated objects (e.g., the pressure on the floor, the force and acceleration impressed to a given object, the tremor of the hands, etc.), and the physiological parameters of the patient when performing that activity (e.g., pulsations, breath, stress, etc.). All the information captured by the sensing system is adequately synchronised and timestamped, and then recorded into a network-attached storage (NAS) composed of 4× 8TB hard drives in ZFS (Zettabyte File System) RAID. At the same time, the saved data are indexed into a MongoDB database for later access by the high-level and presentation layers. In this section, we review the sensors adopted in the system to capture the relevant information.

5.1. IP cameras

A big set of metrics of the proposed expert system are being extracted from simple video streams from common AXIS IP cameras.³ Once a set of simple IP cameras is installed for each environment in the living lab, the proposed expert system takes care of:

- Synchronisation, simultaneously distributing the RTSP stream processing through the nearest computing units
- Video acceleration on GPU using Nvidia CUDA, whenever possible, to enable fast multi-view processing and saving
- Automatic views handoff through human pose estimation, to limit the video recordings to the moments in which the user is interacting with the environment (Fig. 2)
- Fast automatic camera network re-calibration through human mesh recovery, as thoroughly described in Garau, De Natale, and Conci (2020)
- Real-time 3D users position based on IP and depth cameras joint calibration (an example of output is shown in Fig. 3)

A big advantage of our system architecture is that thanks to the fast and fully automatic camera network calibration module, each environment can be easily and dynamically re-configured, without the need of an expert technician for calibration or re-calibration and neither calibration instruments such as checkerboards. The system only requires walking humans in order to re-calibrate itself. Consequently, cameras could be even plugged-in, removed or moved between test sessions, without losing functionalities in the entire system.

Additionally, having a continuously calibrated system of both 2D and 3D vision sensors renders the environment an excellent test bench for both research and working conditions.

Room	Start	End	Duration
	16:49:05	17:00:34	0 hours 11 minutes 28 seconds
	17:00:34	17:05:50	0 hours 05 minutes 16 seconds
	17:05:50	17:09:00	0 hours 03 minutes 10 seconds
	17:09:00	17:42:18	0 hours 33 minutes 18 seconds

Fig. 2. Automatic videos handoff and cutting through global user position monitoring.

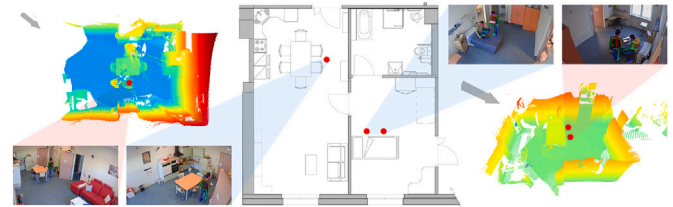


Fig. 3. 3D users position detection through IP and RGBD cameras joint calibration.

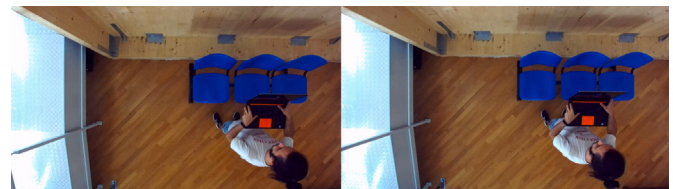


Fig. 4. An example of stereo frame extracted from a ZED 2 camera inside the AUSILIA environment, for fine hands motor skills evaluation.

5.2. ZED cameras

The proposed expert system also allows for the capture and recording of streams coming from more complex cameras, such as stereo cameras. As an example, in the AUSILIA gym environment we employ several Stereolabs ZED⁴ and ZED 2⁵ cameras that allow the system to obtain additional useful information, such as depth maps, high frame rate and resolution videos, refined 3D human poses and meshes. In Fig. 4 you can see an example of the output of a top-view ZED 2 stereo camera inside the AUSILIA gym, used for fine motor skills evaluation.

5.3. RGBD cameras

A set of Microsoft Kinect V2⁶ Time of Flight (ToF) cameras is installed in the domotic apartment to monitor the users' actions and interactions with the objects in the environment. Each Kinect V2 sensor is assigned to a processing unit, which takes care of the image processing and streaming aspects. The configuration of the Kinect multi-camera system ensures a coverage of most of the monitored area. 3D data information is integrated with 2D data to track user position in the environment. Kinect cameras can detect the patient's silhouette starting from the acquired point cloud. To avoid the recognition of unwanted objects in the scene, it is necessary to undergo an initial calibration phase, in which a fused 3D background point-cloud snapshot is saved

⁴ Stereolabs ZED stereo camera <https://www.stereolabs.com/zed/>.

⁵ Stereolabs ZED 2 stereo camera <https://www.stereolabs.com/zed-2/>.

⁶ Kinect V2 Time of Flight camera <https://developer.microsoft.com/en-us/windows/kinect/>.

³ AXIS M1125 IP camera <https://www.axis.com/products/axis-m1125>.



Fig. 5. Virtual bathroom environment with 2 zones where viewer can move.

in order to be subtracted later on to isolate the human silhouettes. Additionally, at each instant of time, an infrared frame is captured by each Kinect V2 in the environment and broadcasted through UDP within the local network. The multiple UDP infrared streams will be used for data processing and fusion, and in particular for the users' identification and tracking, as explained in Section 6 (see Fig. 5).

5.4. Environment and object-mounted sensors

In addition to patient tracking, an evaluation of the user residual manual skills is provided by the analysis of tremors and trajectories during daily activities (brushing teeth, drinking etc.) using LP-Research accelerometers fastened to different objects (toothbrush, glass, etc.). Each room in the domotic apartment is supplied by a set of specific sensors according to the different activities which usually take place in a certain room (Table 1). In particular, in the bathroom six Adafruit Water Flow sensors are installed to detect water flow distribution through the shower, bathtub and sink (measuring both cold and hot water flow in ml/s). At the base of the sink Sa.Ni., *Ultraseensor* pressure matrices can detect the distribution of patient's weight while performing standing tasks. Furthermore, six FSR 400 Interlink local pressure sensors installed under the seat, monitor movements and stability of the patient while standing on the toilet board (Fig. 6).

The kitchen and the living room have different sensors installed: five SeeedStudio Non-Invasive AC Current Sensor (15 A) are responsible for measuring the level of electric current flowing through the household appliances (oven, microwave, induction hob, dishwasher and television) during different user activities. The data relative to the force applied by the user on a table are given by four load cells fastened on each table leg.

In the bedroom, two sensors are placed for the detection of the user in bed. The first is a 100 kg type S Load Cell which is responsible for measuring the force applied by the user when grabbing the trapeze bar over the bed. The second is a SeeedStudio Non-Invasive AC Current Sensor (15 A) which is fastened on the motor of the adjustable electric bed to measure the current flow.

5.5. Biometric sensors

A set of wearable sensors are fastened to patients' body to obtain continuous monitoring of physiological data. Breath rate and one lead ECG are detected using a sensorised Smartex T-shirt while the Blood Volume Pulse (BVP) and Electrodermal activity (EDA) raw data are measured with the Empatica wristband device. A Myo sensor provides measurements on the muscular activity of the forearm, which are useful for understanding the capabilities of the patient in manipulating

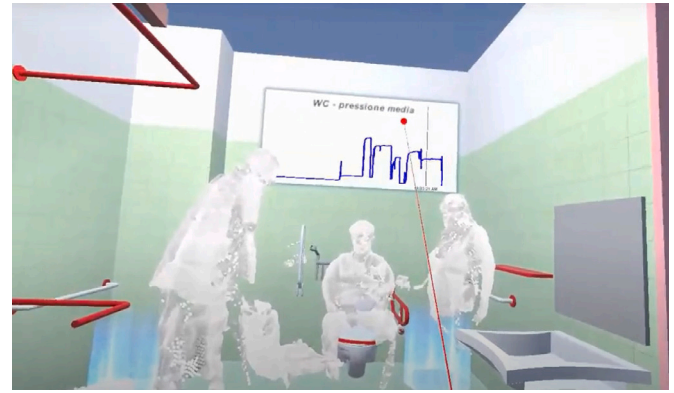


Fig. 6. Example of ambient sensors output: average force applied on the sensorised WC and point clouds of patient and therapists.

objects. Table 2 shows in detail the type of wearable sensors used to monitor the physiological signals and the respective units and sampling frequencies. Wearable sensors allow a non-invasive detection of the physiological parameters, but it is necessary to be careful while applying these devices to the patient. The sensorised T-Shirt should be the right fit, in order to avoid loss of signal due to a wider size or an uncomfortable sensation due to a tighter size.

6. Processing subsystem

We have seen how the highly configurable properties of the proposed system allows for fast and structured data collection by sensors and cameras across all the environments. However, many times raw data signals may need to be filtered and re-organised according to the specifications of the task. For this reason, it is possible to rely on the underlying MQTT broker to write custom modules that take as input raw sensors data, process it and return custom outputs that are going to be saved as a new record in the database and in the attached storage. Some use cases of custom data processing modules implemented inside the AUSILIA domotic apartment are presented below.

6.1. Users identification and tracking

In ambient-assisted living scenarios, many metrics derive from the interactions of the user with the environment. Therefore, it is of fundamental importance to be able to identify and track users and caregivers separately. Commercial indoor tracking devices exist for that purpose, but they may not be ideal to adopt in certain scenarios, such as:

- Large coverage needs
- Complex indoor topology
- Need for non-intrusive technology
- Restricted budget

Additionally, having a complex and already calibrated camera network at disposal makes it easier to build a vision-based identification and tracking system.

We propose a vision-based approach for users and caregivers identification and tracking, which relies both on a real-time people detection network working on infrared images (based on [Bochkovskiy, Wang, & Liao, 2020](#)) and a simple tracking-by-detection algorithm. An overview of how the proposed detection pipeline can be seen in algorithm 1. To sum it up, we take as input the infrared images I from the Kinect V2 sensors, streamed via UDP, and the set of camera matrices K . We apply an off-the-shelf infrared people detector (Fig. 7) and project the detection on to the global point cloud of the environment to obtain a 3D coordinate for each detected person. Subsequently, we apply a threshold on the infrared image, to detect specific reflective patterns

Table 1

Details and specifications of the employed environmental sensors.

Sensor	Description	Signal	Unit	Advantages	Disadvantages
Camera	Microsoft Kinect V2, Time of Flight (ToF) camera. It provides 30 Hz data of RGB, infrared and depth images	2D image	/	– Low cost	– No trigger for multiple device synchronisation – Uncertainty of about 5 mm at 5 m
		Infrared image	/	– Automatic skeletonisation of people moving inside its field of view	
		Depth image	/	– Markerless	
IMU	Lp-Research “LPMS-B2”	3-axis magnetometer	μT	– Low cost	– Error that accumulates over time
		3-axis accelerometer	dps	– Small size ($39 \times 39 \times 8$ mm)	
		3-axis gyroscope	ms^2	– High data transmission rate: 400 Hz	
Pressure matrix	Sa.Ni. Corporate “Ultrasensor” installed in front of the sink.	Force	N	– Low cost – Large measuring area (0.5×0.5 m) with a spatial resolution of 42 dpi	– It requires custom built supports
Current flow	Installed to evaluate the correct use of electrical devices in the apartment	Current	mA		– Difficult to assemble
Water flow	Installed to evaluate the correct use of the cold and hot water by the user	Water flow	ml/s	– Large temperature range: from -20°C to 80°C – Large flow rate: from 1 Lmin^{-1} to 30 Lmin^{-1}	– Difficult to assemble
Load sensor	Interlink “FSR 400 Series” installed under the table in the kitchen, under the toilet seat and on the bar next to the toilet	Force	mg	– Low cost – Easy to configure	– It requires custom built supports

Table 2

Details and specifications of the employed wearable sensors.

Sensor	Description	Signal	Unit	Sampling frequency [Hz]	Advantages	Disadvantages
T-shirt	Wearables Wellness System (WWS) T-shirt by Smartex. It provides the ECG signal (lead II) and the respiratory signal	ECG	mV	250	– Easy to integrate with other sensors – Programmable hardware – It does not require the application of electrodes	– T-shirt must be the right size – Electrodes might be very sensitive to movements if the T-shirt is not worn correctly
		Breath rate	mV	25		
		3-axis accelerometer	ms^2	25		
Wristband	Empatica E4 wristband used to record the BVP signal through its photoplethysmographic (PPG) sensor. The wristband provides also the EDA signal and the skin temperature.	BPV	nW	64	– It contains more sensors than a normal wristband device – Possibility to obtain high-quality raw sensor data in real-time – It does not require the application of electrodes	– It should be worn reasonably tight to correctly detect the signal – It needs specific software to access raw data
		EDA	μS	4		
		3-axis accelerometer	ms^2	32		

Algorithm 1: Find a non-repeating set P of people containing their global 3D position and a label that refers to a caregiver (0) or a patient (1)

PatientDetector (I, K)

inputs: A set $I = i_0 \dots i_n$ of infrared images; a set $K = k_0 \dots k_n$ of camera matrices
output: The labelled 3D positions $P \mid p_i \in P$ of the user and caregivers, where $p_i = x, y, z, \text{label}$

```

 $P^* \leftarrow \emptyset;$ 
 $P^* \leftarrow \text{InfraredDetector}(I);$ 
foreach person  $p_i \in P^*$  do
   $t_{p_i} \leftarrow \text{threshold}(p_i);$ 
  if  $t_{p_i} \geq 0$  then
     $p_i[\text{label}] \leftarrow 0;$ 
  else
     $p_i[\text{label}] \leftarrow 1;$ 
 $P \leftarrow \text{merge}(P, K);$ 
return  $P;$ 

```

that can be found on the occupational therapists' uniforms. By doing so, we can assign a label 0 (caregiver) or 1 (user) to each coordinates triplet p_i . Finally, we obtain the set of unique labelled 3D positions P and return it as output. Algorithm 1 is repeated for each instant of time.



Fig. 7. People identification through IR (infrared) sensors (left) and tracking-by-detection (right).

6.2. Motion trajectories extraction

Human trajectories are of extreme importance in computer vision. Research fields such as trajectory prediction count increasing numbers of publications in literature and trajectory datasets are ever more requested. However, annotating a trajectory dataset is still a cumbersome process, that needs to be supervised in many cases, mostly due to occlusion and re-identification issues. Having a calibrated multi-camera system at disposal makes it easier to automatically annotate trajectories. As an example, in the AUSILIA gym we apply a mixed background subtraction and human pose estimation in order to obtain filtered and smooth trajectories, even detecting spatial and temporal breakpoints.



Fig. 8. Raw user trajectory data points (green) (left, apartment) and final trajectory with metrics (right, gym).

Further details about the process are explained in Sebastiani, Garau, De Natale, and Conci (2019) and an example of a resulting trajectory can be seen in Fig. 8.

6.3. Occupancy heat-map generation and habits tracking

Often times, occupational therapists have to scout through up to hours of video footage in order to estimate the occupation of environments by certain users. This approach is indeed not optimal in terms of time optimisation, and it is prone to errors. As an alternative, heatmaps can be built starting from raw positional data. As an example, in Fig. 9 a heatmap of the daily occupation of the AUSILIA domotic apartment is shown. Other possible visualisation can be built from this type of data, such as heatmap representing interactions between patients and caregivers, or danger zones for social distancing, which in these days is a hot topic indeed (Lewnard & Lo, 2020).

By combining both the room occupancy and the wearable and environmental sensor data, it is possible to keep track of the users' habits, and even automatically detect and label them once they occur. Many other works, such as Alemdar, Ertan, Incel, and Ersoy (2013), tackle this problem, although with some manual intervention for labelling.

6.4. Sensor data integration

User identification, location, and tracking information are integrated with other wearable and environmental sensor data. All collected data, previously filtered, recorded, and synchronised, allow a final presentation with all the information acquired. It provides a detailed overview of all the needed information to better estimate user activities. These notions are then displayed simultaneously through a dedicated web page and an immersive interface.

7. Presentation subsystem

From a data presentation viewpoint, the proposed expert system provides three possible levels of access to different type of data, namely:

- Debugging sensors and modules signal through the system management console
- Accessing quantitative data, as well as video streams and metrics for each environment through a web-based dashboard
- Accessing qualitative data, in the form of 3D reconstructions of the environment and users actions, through the usage of a HTC Vive⁷ head-mounted display (HMD).

Every level of data access is further detailed in the following sections.

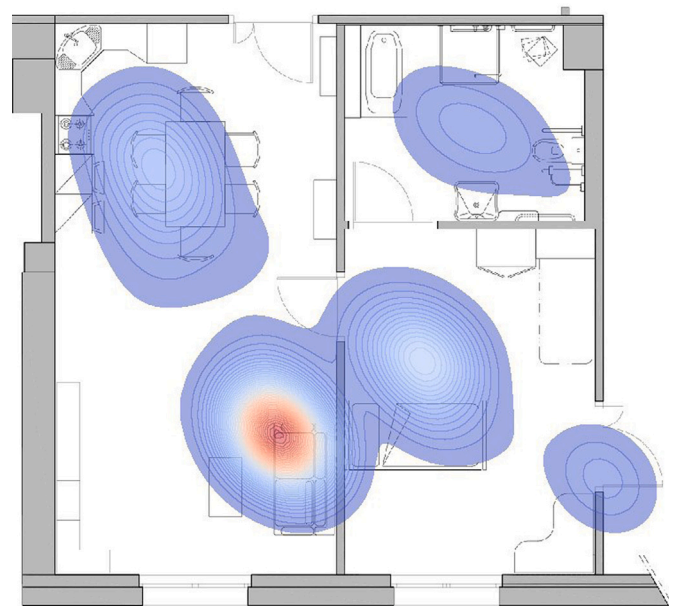


Fig. 9. An example of the generated occupancy heat-map for the AUSILIA domotic apartment.

7.1. System management console

Each room and scenario of the AAL environment can be accessed through a system management console, which has been written using the Unity game engine, to maximise its portability through devices and operating systems, including WebGL frames. This makes it possible to run the system management console even from a browser or a simple smartphone device. The main purpose of the console is to check the status of each sensor and camera inside the environment, as well as providing additional crafted functionalities, such as preventing the usage of an environment if there are ongoing tests or if critical components are not functional or not responding. An example of the system management console dashboards can be seen in Fig. 10.

7.2. Web-based data presentation

While the system management console can be used to check the status of the entire system, including sensors and cameras for each environment, the web-based interface can be used to access data coming from the different tests, for each user in the system (Fig. 11). Each member of the medical staff can access, through a login screen, to different pools of patients, based on their assignments. Patient data can be shared among different members of the medical staff if required. An example of possible output data that the users have access to while navigating the web dashboard is reported in Table 3.

7.3. HMD-based data presentation

The HMD used by the therapists can be seen as an enhanced display screen in front of their eyes. This device allows them to enter a digital world and fully immerse themselves in it with all the information need to increase the objectivity and effectiveness of their clinical observation.

From the scale of Mixed Reality (MR), which refers to the fusion of the real and virtual world, our context is in Augmented Virtuality (AV) field. The patient's real data and the therapists themselves are projected

⁷ Htc Vive head-mounted display <https://www.vive.com/us/>.

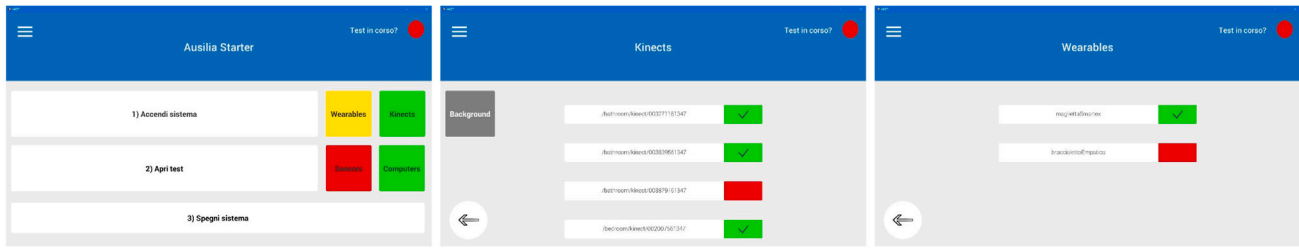


Fig. 10. Example screenshots from the Unity dashboard to turn on/off the system and its components.

Table 3

Example of possible output data that can be visualised through the web-based dashboard.

Datum	Explanation
R_p	The patients' p registry
T_p	A list of tests for each patient p
$E_{p,t}$	A list of environments used for each test t
$V_{p,e,t}$	A list of video sequences for each utilised environment e in each test t
$S_{p,e,t}$	A list of sensor data sequences (wearable and environmental) for each environment e in each test t
$H_{p,e,t}$	A occupational heatmap for each environment e in each test t
Other	Any kind of data that requires additional visualisation

Table 4

Actors and their roles in the conducted experiments.

	Description	Interacts with
P	The patient P is assisted, if needed, by the occupational therapist OT1 during its stay inside the domotic apartment DA.	[OT1, DA]
OT1	The occupational therapist OT1 is the only person that can directly assist the patient P inside the domotic apartment DA.	[P, DA]
OT2	The occupational therapist OT2 has a blind role, meaning that their only knowledge of the patient P is provided by the expert system ES.	[ES]
DA	The domotic apartment DA hosts the patient P and the occupational therapist OT1, simultaneously interfacing with the expert system ES.	[P, OT1, ES]
ES	The expert system ES interfaces with the domotic apartment DA and the occupational therapist OT2, collecting and organising information from the former, which can be then accessed by the latter	[OT2, DA]

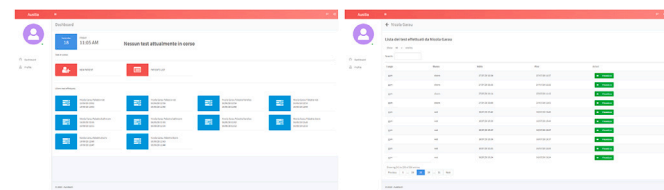


Fig. 11. Home screen and list of tests from the web dashboard.

into a virtual world and participate in it. The virtual environment is a simulated model of the real environment.

The main advantage of AV lies in the possibility for patients to be independent and for therapists to assess the patients' status from an augmented view. Indeed, it avoids the possibility of distraction and influencing the patient's behaviour due to the presence of an external observer. Furthermore, in addition to the evidence provided by the human eye, a large amount of data can be provided to the therapist.

The therapist's augmentation can be related to the patient's motion in 3D, their interaction with the environment and their physiological parameters with the data acquired by the multiple 3D cameras, the different environmental sensors and the wearable devices we discussed in Section 4.

The use of innovative technologies using MR methodologies, once the device and what parameters of interest to show were selected, make it necessary to provide data with appropriate feedback to the final user. Only an intuitive, effective and user-friendly interface makes these technologies a useful tool, without affecting the therapist's mental effort. Therapists have control of the entire virtual scene aided by the use of the HTC Vive controller. In addition, it is possible to navigate within the environment with natural gestures (e.g. looking around, moving, etc.). In particular, they can decide for each action of the patient, which parameters to display, at what time and from what position and point of view.

In the 3D domain, navigation is much more effective in terms of speed and truthfulness. It is possible to directly estimate the 3D shape of objects in the scene and link the corresponding information. Contextualising the measured quantities with the area in which these data were collected allows easy interpretation due to the natural link between cause and effect. Some data are displayed through 2D panels within the scene with a pre-set graphic style. These panels contain static diagrams but with the possibility to enables exploration over time.

The medium used to provide information is not only related to visual feedbacks, but further possibilities are 3D sound recreated for the patient's heartbeat and breath or haptic feedback driven by the 3D interaction between the controller and the objects displayed in the virtual scene. The combination of these stimuli suggests to the therapists a natural correlation of different quantities with respect to the patient's status.

8. Experimental results

Objectively assessing the patients performances solely on the basis of pre-defined metrics (e.g. room occupancy, required OTs assistance) is not a common practice, because each patient presents different needs and variable degrees of independence, to name a few. For this reason, we argue that testing expert systems, such as the ones proposed in this work, requires additional supervision by the experts, namely the medical personnel (in this case mainly doctors and OTs).

8.1. Living lab setup and experiments configuration

In order to assess the capabilities of the presented expert system, we designed a series of experiments which involved different actors, each presenting a different role and set of interactions, as shown in Table 4.

All the tests have been conducted in a blind way, meaning that only one of the two occupational therapists (OT1) was allowed to interact with the patient P inside the domotic apartment DA, while the

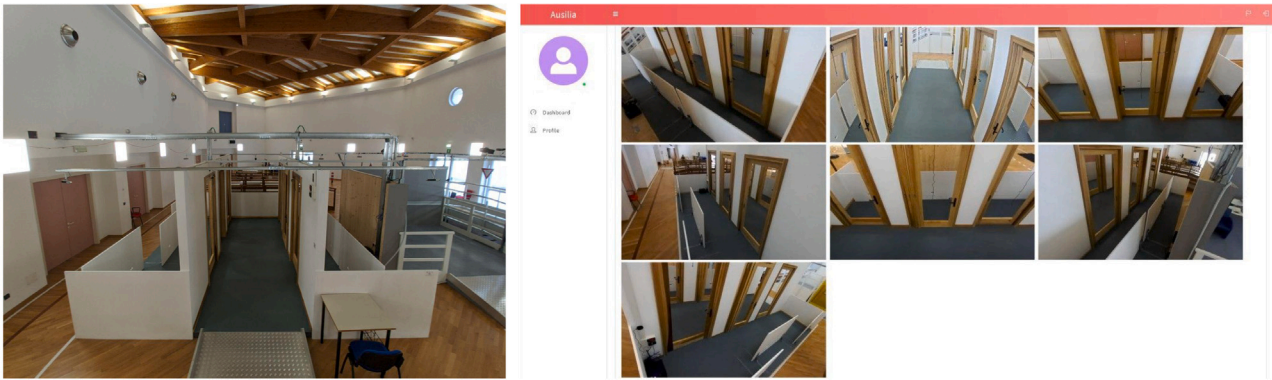


Fig. 12. Ausilia's Doors environment.

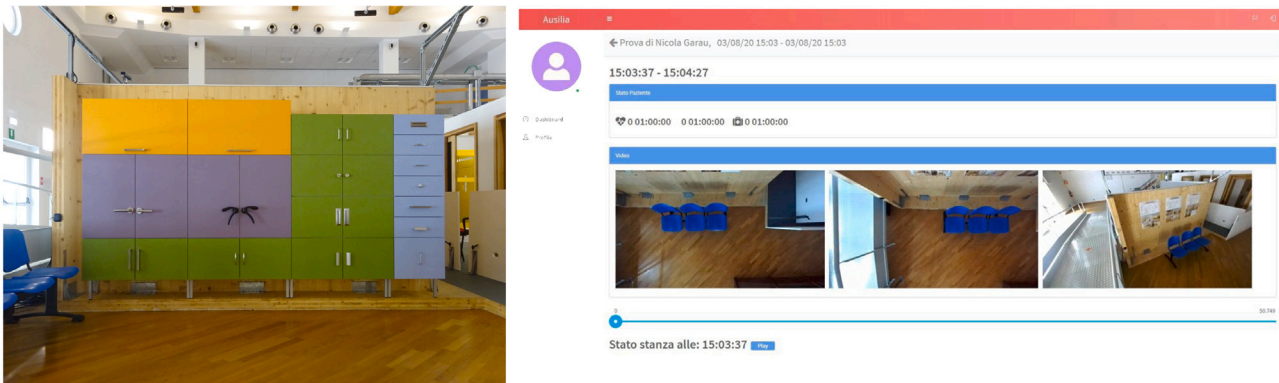


Fig. 13. Ausilia's Handles environment.

other (OT2) had to stay outside the whole environment, and monitor everything by only using the web dashboard and the 3D virtual reality visualisation provided by our expert system *ES*. At the end of each test, they were required to fill a standard report describing the performances of the user for each test. The desired outcome of the conducted test was to minimise the distance between the evaluations of patient *P* by the occupational therapists *OT1* and *OT2*, despite the drastically different knowledge sources (directly provided by [*P*, *DA*] for *OT1*, indirectly provided by *ES* for *OT2*).

8.2. Preliminary results

Results showed that the proposed expert system can be of great help in aiding the medical staff to objectively evaluate the patients' performances. Moreover, having the possibility of re-evaluating tests after their completion in an offline fashion made it possible for *OT2* to take into considerations some little behavioural details that *OT1* was unable to grasp in person.

A very important factor of the possibility of remotely evaluating patients performances is fostering the complete independence of the user inside the AAL environment, without any help from any occupational therapist, thus accurately reproducing a realistic standard home scenario.

8.3. Data collection and quantitative results

The proposed expert system has been used inside and tailored for the AUSILIA project (Pisoni et al., 2016), inside the Villa Rosa hospital in Pergine, Italy. During the past years, we worked alongside the medical personnel in order to refine the required metrics, inputs and outputs. During the first year of the system's experimental phase, 52 volunteer patients have been interacting with the system. Each patient

stayed in the AAL environment for a total of 2–3 days, mainly during daytime day hospital hours. For this reason, most of the data has been recorded in the living room, kitchen and bathroom. As a result, until today the presented expert system was able to automatically collect and annotate the following data, that is available for consultation by the medical personnel for additional analysis:

- Over 5k video sequences, annotated by zone, test type and patient present in the video
- Over 5k pose and trajectory sequences (Fig. 8), for both the patients and for the OTs
- Corresponding occupancy maps (Fig. 9) for each conducted test
- More than 5k annotated biometric and environmental data sequence
- A 3D point cloud sequence for each conducted test

In total, approximately 5 terabytes of data have been collected during the past year of testing. All the past tests can be accessed at any time by the authorised medical personnel through the web dashboard, in order to study, evaluate and re-evaluate certain conditions through quantitative metrics that were previously impossible to objectively verify. The presented expert system can thus be used as an additional occupational therapy tool for understanding how to provide the best possible help to new patients, based on previous similar cases.

8.4. Scalability and replicability

The proposed expert system is currently adopted by the AUSILIA⁸ living lab (TN, Italy). The AUSILIA's apartment is currently fully operational and self-sustaining. Similarly, the AUSILIA's gym (Figs. 12,

⁸ Project AUSILIA (Assisted Unit for Simulating Independent Living Activities) <http://ausilia.tn.it>.

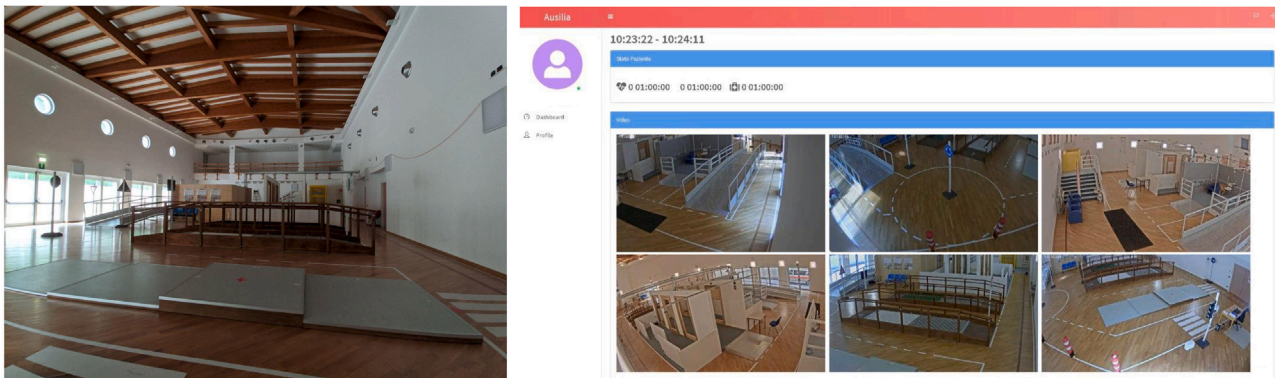


Fig. 14. Ausilia's WST environment.

13, 14) has been also finalised from both an architectural and sensorial viewpoint.

It is worth recalling that the AUSILIA framework is scalable by design, and it is conceived to allow the integration of new modules, based on the environment and clinical requirements, also allowing for the deployment of only a subset of technologies and analytics, upon need.

The proposed systems is fully replicable: the hardware setup mostly consists of general-purpose off-the-shelf components, widely available on the market. The software architecture is built upon established open-source frameworks and libraries. Its customisation is required to suit the needs of the specific physical environments, which maps the clinical guidelines and thus follows strict criteria in terms of architecture, design, and ergonomics. The following points have to be carefully taken into account:

- The location should be carefully selected to guarantee safe and controlled access to patients and medical doctors.
- The environment should be largely reconfigurable: fixed setups would not allow generating custom settings for a specific patient that are both realistic and challenging. This is crucial in ensuring an effective training simulation.
- It might be sufficient to deploy a selected subset of the home environments. For instance, kitchen and bathroom often turn out to be the most critical and potentially dangerous zones, and may already provide sufficient information for many user studies.
- The sensing infrastructure should be carefully planned according to the goals of the analysis: a preliminary study with medical doctors is necessary to identify the significant parameters to be extracted, and to select the most appropriate sensors accordingly (tables provided in Section 5 could be used as a reference).
- Data visualisation tools are of paramount importance for the therapists: the quantity of data collected by the system is overwhelming, therefore it is necessary to select, contextualise and simplify the generated information flow through appropriate interfaces and devices.

9. Conclusions

We presented an expert system targeting ambient-assisted living scenarios, which is fully automatic and capable of monitoring, recording, annotating and storing the users' daily activities. The performance level of each activity is expressed in the form of objective metrics, automatically calculated from a combination of raw video and sensor data. The metrics are specifically tailored to target the pre-filling of standard user cards, which the OTs are usually required to compile by hand. The proposed expert system allows therefore to significantly reduce the patients' performance evaluation efforts. At the same time, it can be used as an assistive technology for helping in prescribing technical aids specifically tailored to the different needs of each patient.

As to the future evolution of the system, we plan to provide additional metrics through different sensors data fusion, as well as a better visualisation of the reconstructed environments. We are currently planning to employ techniques similar to VIBE (Kocabas, Athanasiou, & Black, 2020) to provide a completely navigable 3D environment to let the medical staff better observe the users action from different viewpoints. A great advantage of using these kinds of techniques is that we could even reinforce the 3D pose accuracy by mixing the ground truth 3D data with the inferred meshes and poses, in a similar fashion to what is described in Hassan, Choutas, Tzionas, and Black (2019).

CRedit authorship contribution statement

Nicola Garau: Conceptualisation, Methodology, Software, Writing – original draft. **Damiano Fruet:** Data Curation, Methodology, Software, Writing – original draft. **Alessandro Luchetti:** Writing – review & editing. **Francesco De Natale:** Supervision, Project administration. **Nicola Conci:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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